Perturbation Evaluation Report

Source file: C:\Users\sachi\Desktop\VLM_Benchmarking\temp\eval_patch_20260407_194711_metrics.json
Generated at: 2026-04-08 01:28:04

Executive Summary

Most vulnerable model: **clip** (risk score 0.2734). Most robust model: **deit_tiny** (robustness 0.9419).
Average ASR: 15.67%
Average ODS: 7.18%
Average SBR: 0.00%

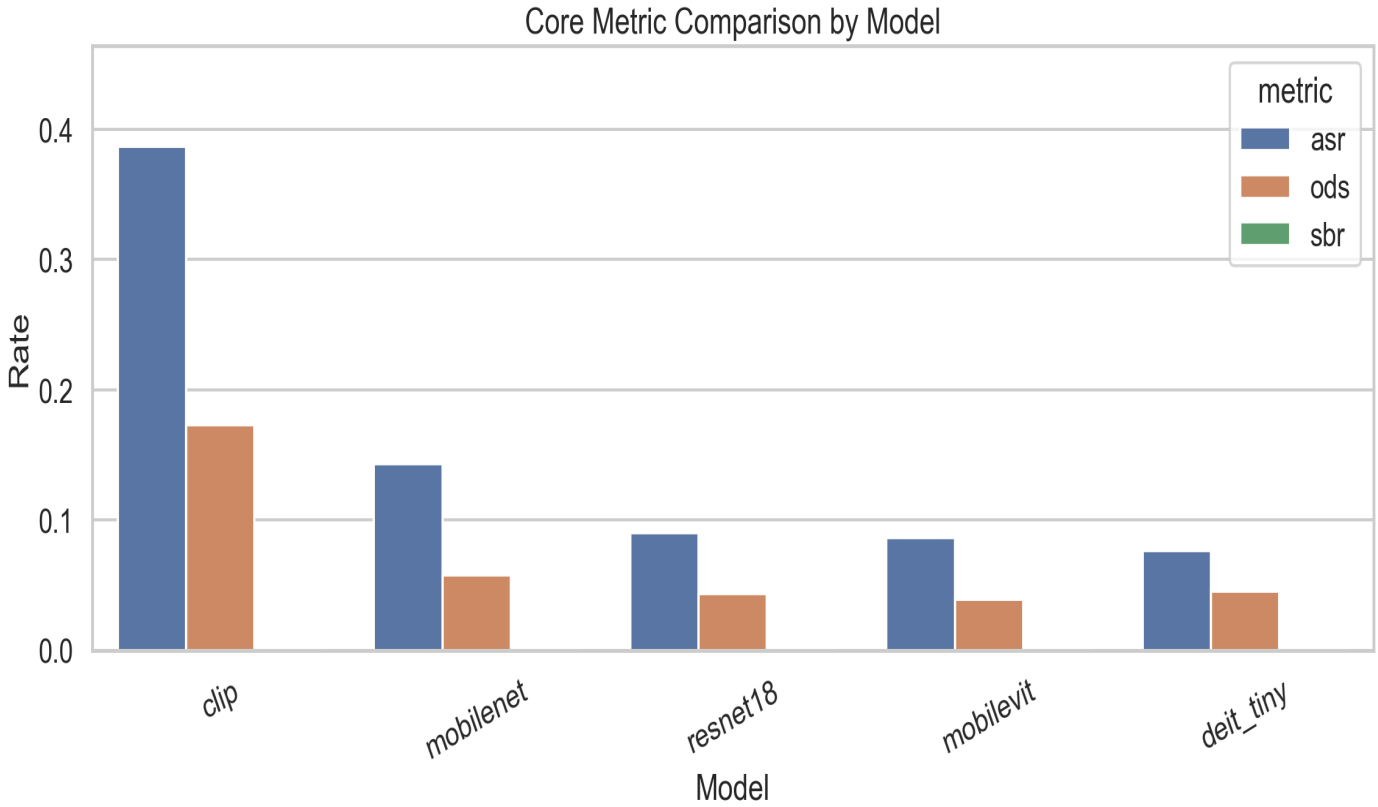
Detailed Metric Table

Rank	Model	Pairs	ASR	ODS	SBR	Risk	Robust	Succ/1000	Drift/1000	Z
1	clip	300	0.3867	0.1734	0.0000	0.2734	0.7266	386.7	173.4	1.968
2	mobilenet	300	0.1433	0.0576	0.0000	0.0990	0.9010	143.3	57.6	-0.149
3	resnet18	300	0.0900	0.0434	0.0000	0.0647	0.9353	90.0	43.4	-0.566
4	mobilevit	300	0.0867	0.0391	0.0000	0.0613	0.9387	86.7	39.1	-0.606
5	deit_tiny	300	0.0767	0.0455	0.0000	0.0581	0.9419	76.7	45.5	-0.646

Visual Analysis

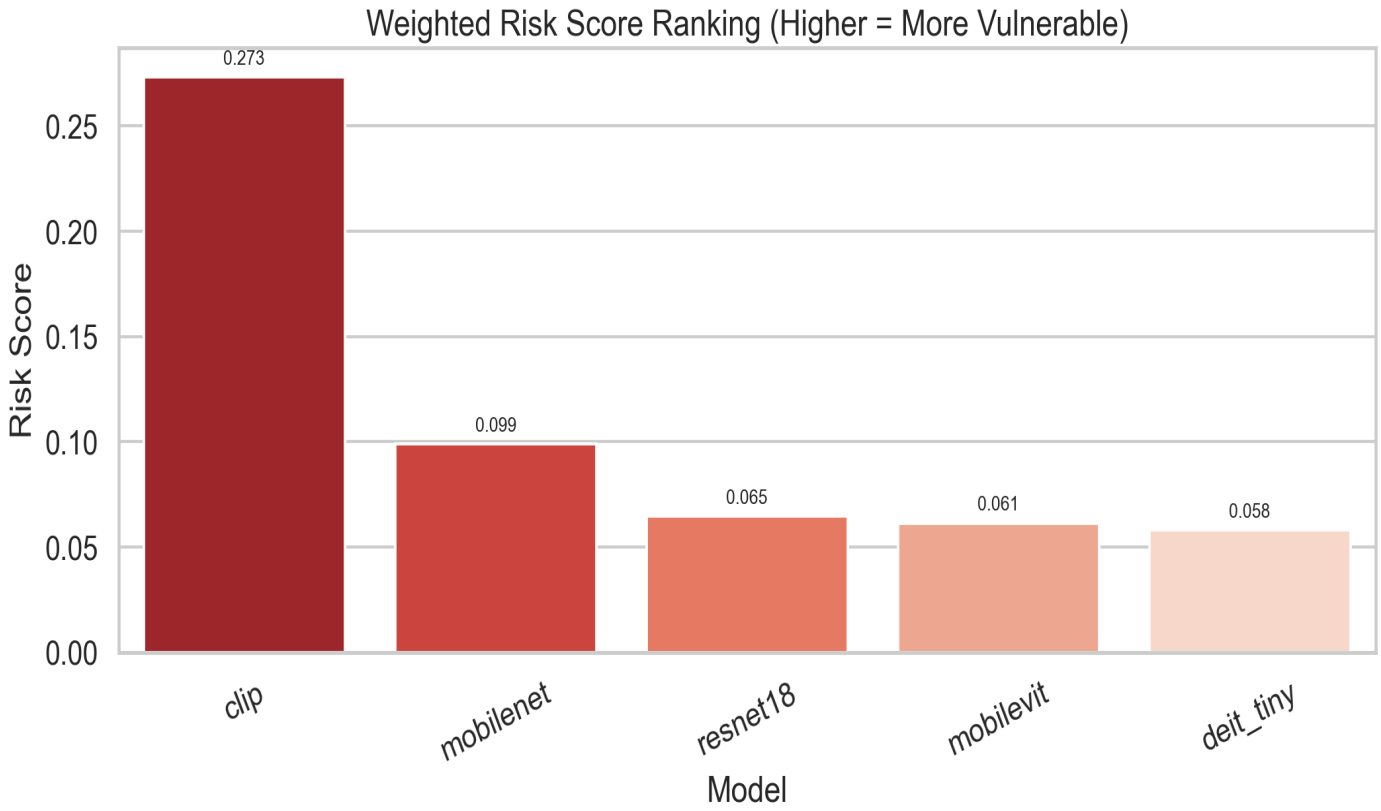
1. Core Metric Comparison

Side-by-side ASR, ODS, and SBR for each model.



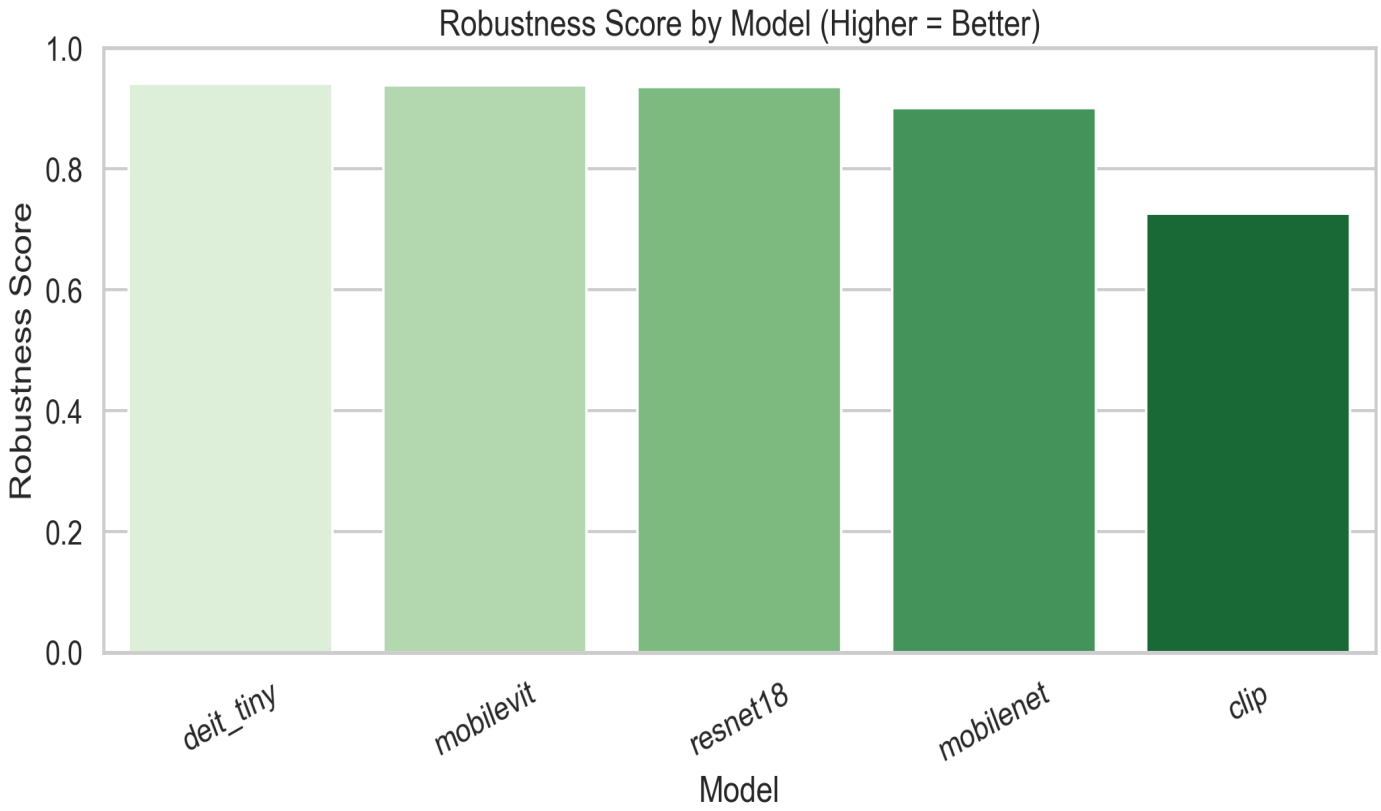
2. Risk Ranking

Composite vulnerability score with weighted metrics.



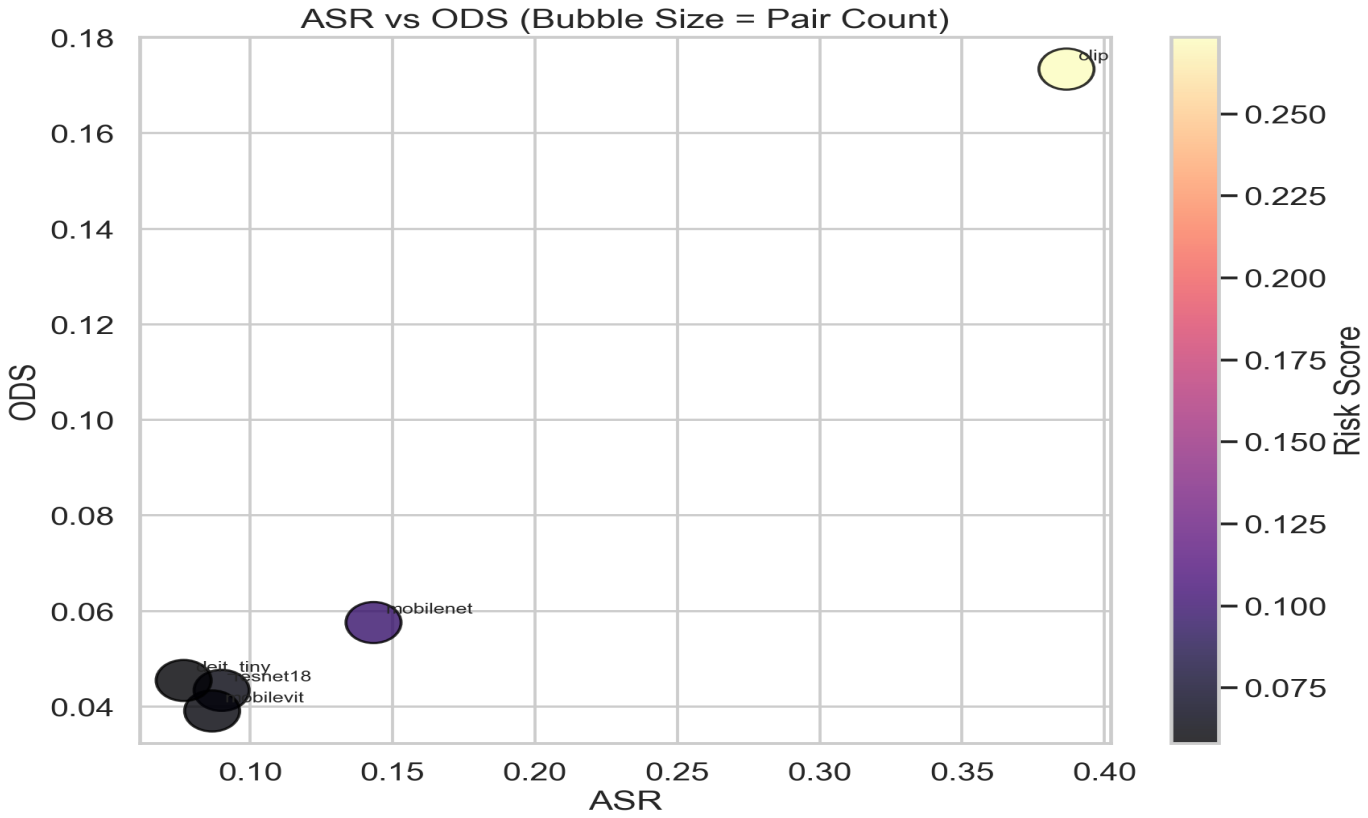
3. Robustness Scores

Inverse of weighted risk for easy model ranking.



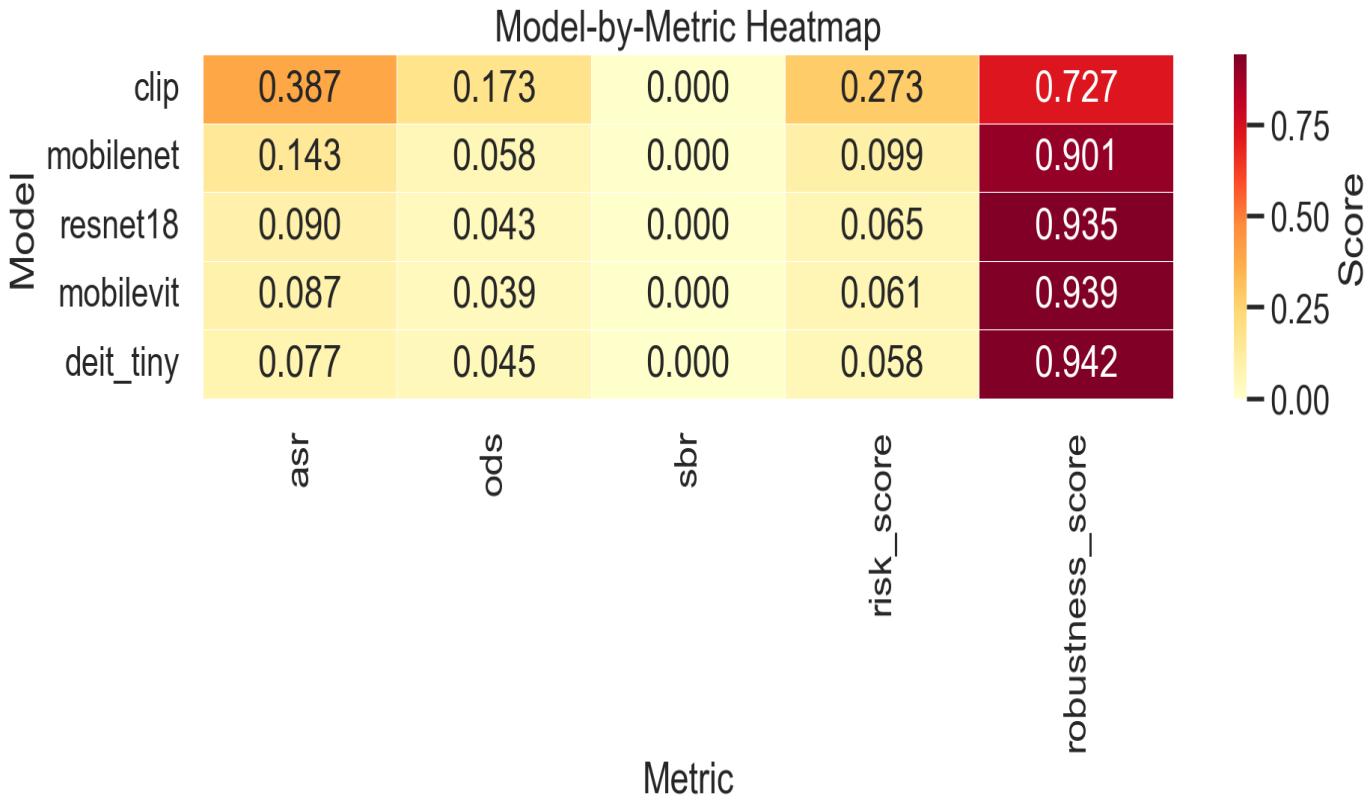
4. ASR-ODS Bubble Plot

Relationship between attack success and output drift.



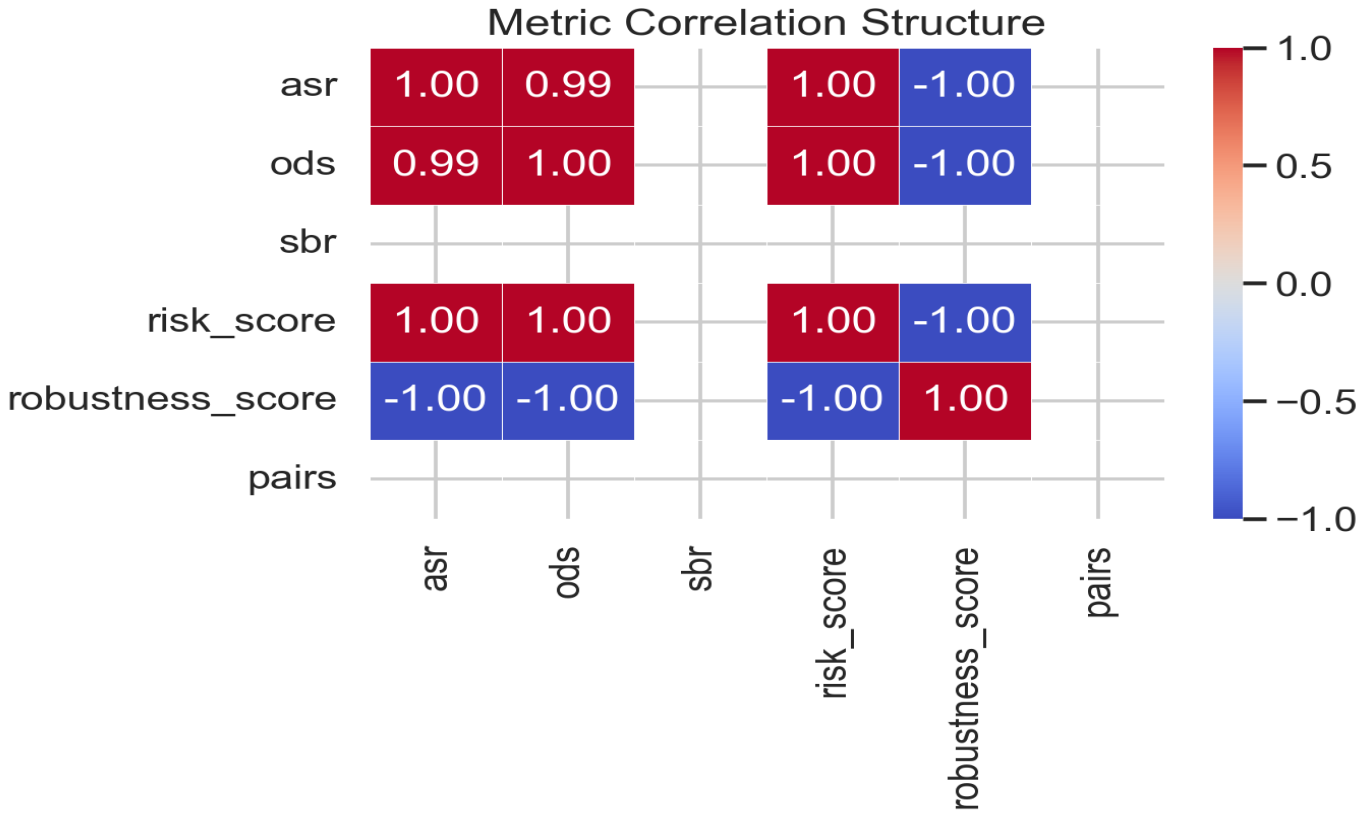
5. Metric Heatmap

Consolidated intensity map across all key metrics.



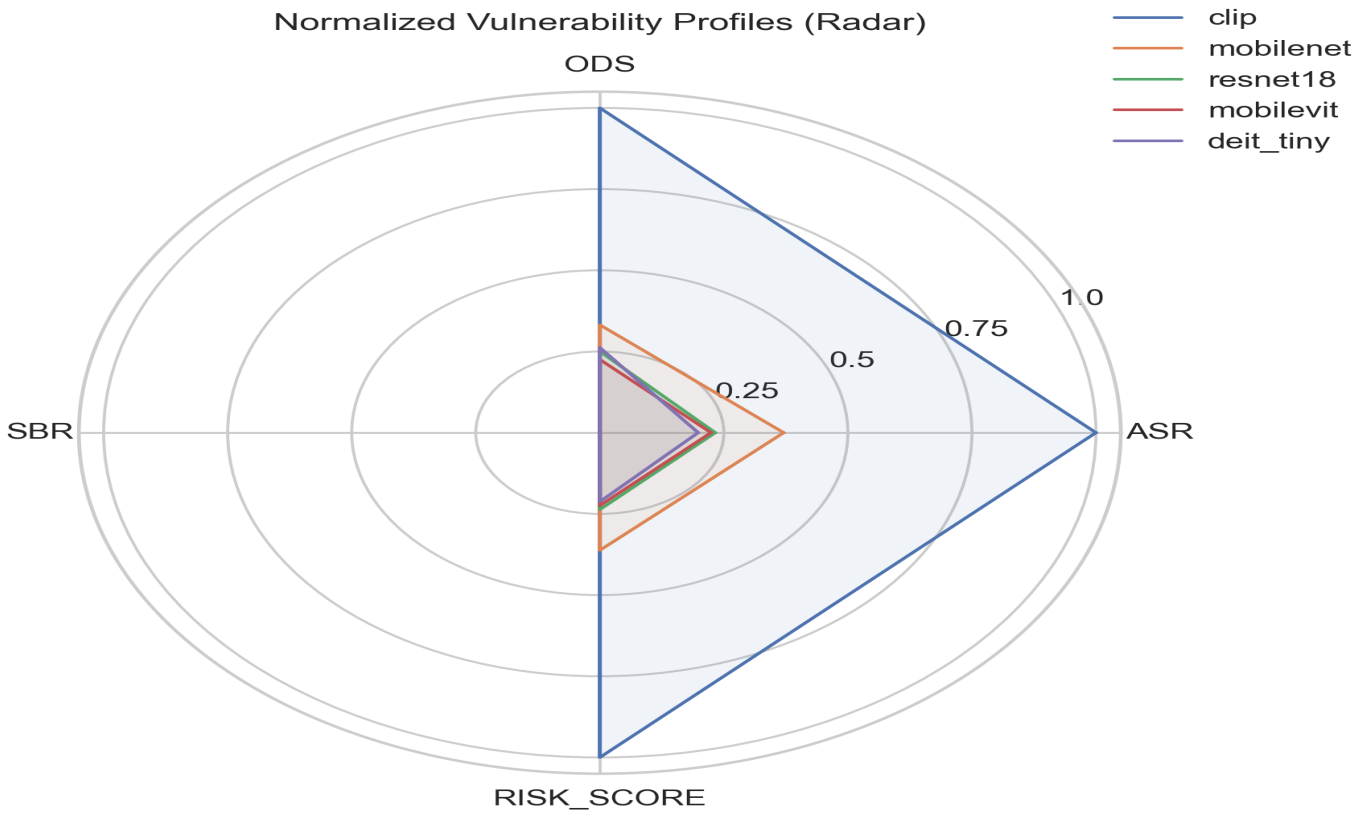
6. Correlation Heatmap

Linear relationships between attack and robustness metrics.



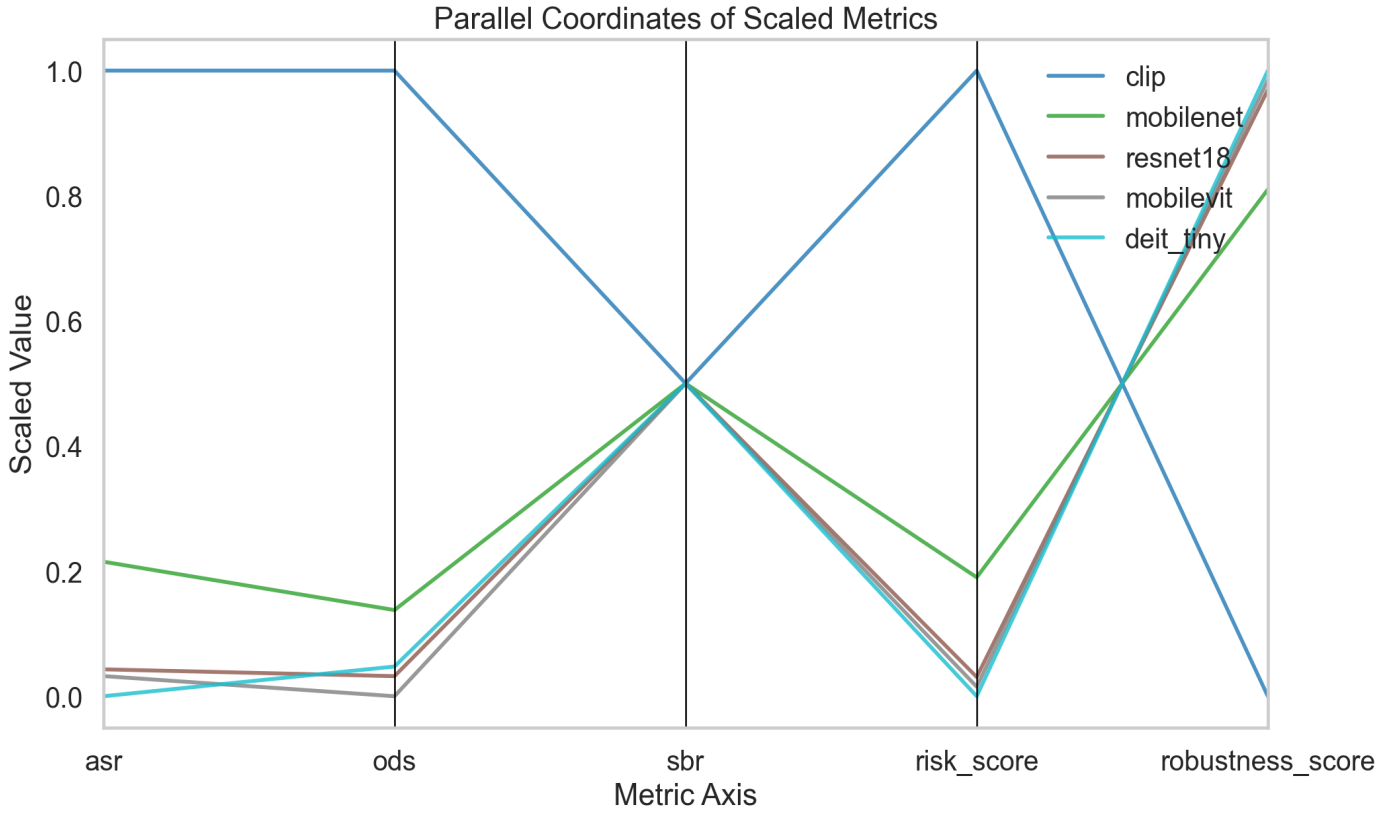
7. Radar Profiles

Normalized per-model vulnerability signature.



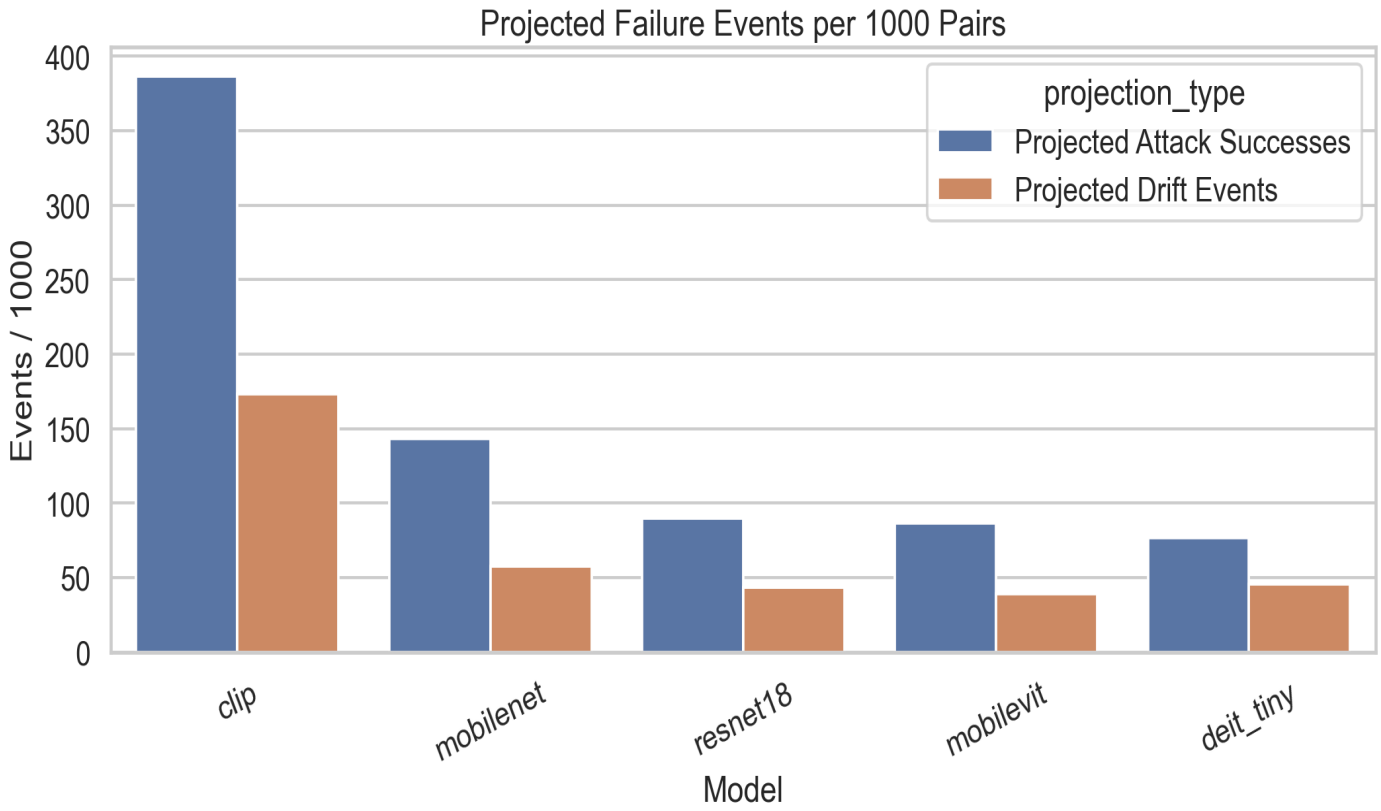
8. Parallel Coordinates

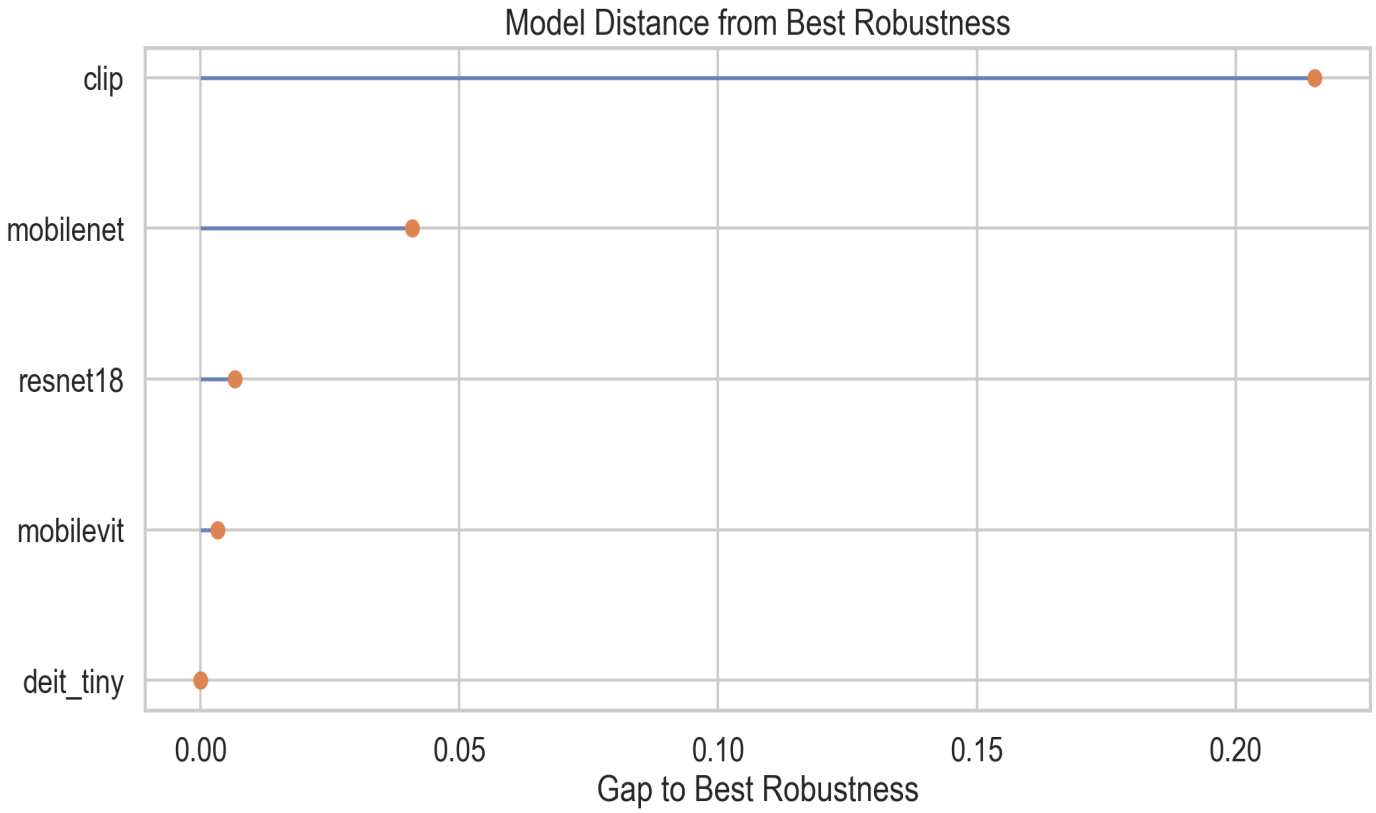
Multi-metric trajectory of each model on a normalized axis.



9. Projected Failure Events

Normalized operational-risk projection per 1000 pairs.



10. Gap-to-Best

How far each model is from the strongest observed robustness.